INCH-POUND
MIL-PRF-3098/90C
w/AMENDMENT 2
4 June 2025

SUPERSEDING
MIL-PRF-3098/90C
w/AMENDMENT 1
24 June 2019

PERFORMANCE SPECIFICATION SHEET

CRYSTAL UNIT, QUARTZ, CR114/U

This specification is approved for use by all Departments and Agencies of the Department of Defense.

INACTIVE FOR NEW DESIGN AFTER 4 JUNE 2025

The requirements for acquiring the product described herein shall consist of this specification sheet and MIL-PRF-3098.

Pertinent characteristics: 2.9 MHz to 3.85 MHz; fundamental; noncontrolled; series resonance.

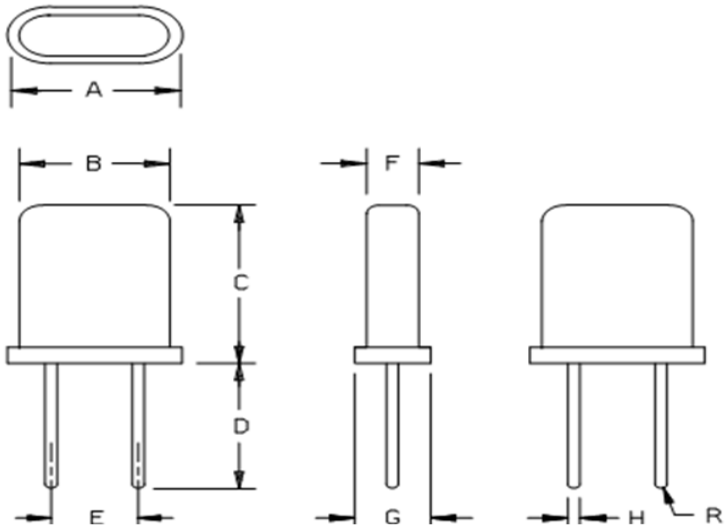


FIGURE 1. Crystal unit - CR114/U.

AMSC N/A
DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

FSC 5955

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Ltr	Inches		mm	
	Min	Max	Min	Max
A	---	.435	---	11.05
B	---	.402	---	10.21
C	---	.530	---	13.46
D	.120	.130	3.05	3.30
E	.184	.200	4.67	5.08
F	---	.150	---	3.81
G	---	.183	---	4.65
H	.038	.042	0.96	1.07

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Marking to be in accordance with MIL-PRF-3098.

FIGURE 1. Crystal unit - CR114/U - Continued.

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REQUIREMENTS:

Dimensions, marking, and configuration: See figure 1.

Frequency range: 2.9 MHz to 3.85 MHz, inclusive.

Frequency tolerance:

Operating temperature range: ±0.005 percent.

Mode of oscillation: Fundamental.

Operating temperature range (noncontrolled): -55°C to +105°C, inclusive.

Capacitance, load: 32.0 pF ±.05 pF.

Resonance: Series.

Rated drive level: 1.0 mW, maximum.

Calibration values:

Resistance: 100 ohms.

Crystal current: 3.2 mA.

Capacitance, shunt: 7 pF, maximum.

Equivalent resistance: 300 ohms, maximum.

Shock: MIL-STD-202-213

Frequency change permitted: ±0.0005 percent.

Equivalent-resistance change permitted: ±15 percent.

Vibration: MIL-STD-202-204, test condition A; .10 inch (2.54 mm) double amplitude or 5 gravity units, whichever is less.

Frequency change permitted: ±0.0005 percent.

Equivalent-resistance change permitted: ±15 percent.

Unwanted modes: Minimum unwanted mode effective resistance shall be at least twice that of the main mode, or greater, over the operating temperature range -55°C to +105°C.

Aging:

Frequency change permitted: ±0.0005 percent.

Part or Identifying Number (PIN): CR114/U (followed by specified frequency), see 1.2 of MIL-PRF-3098.

Qualification inspection is not applicable for this specification sheet. Acceptance testing shall consist of screening, Group A and Group B inspection of MIL-PRF-3098.

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Reference documents. In addition to MIL-PRF-3098, this document references the following:

MIL-STD-202-204
MIL-STD-202-213

Amendment notations. The margins of this specification sheet are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

Custodians:
Army - CR
Navy - EC
Air Force - 85
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Review activities:
Air Force - 19

Preparing activity:
DLA - CC

(Project 5955-2025-007)

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST Online database at https://assist.dla.mil/.

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